Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.8549598323436954	0.8378494623655914 – 0.8691263176701804

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.3604773282010496	0.35471006182411524 – 0.38053385725526634

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.44287614487737254	0.4398298429214799 – 0.4580180849626527

Thin QR

512 × 128 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.7576909363946751	0.7432072422955074 – 0.8038720198704554

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.4991791918264273	0.4803599515307621 – 0.5078611944426149

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9741677808632452	0.9438496742229416 – 0.9984214886844286

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9990550437042287	0.9766783174641437 – 1.0745344105206183

Thin QR

256 × 64 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9788347720307858	0.9634141448186757 – 0.989606890943739

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0133164235890932	0.9890243902439024 – 1.1006211180124224

Matrix multiplication

128 × 128 · 128 × 128 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9941706698332163	0.9845926337875073 – 0.9993919658419922

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0183357601251146	0.993922368385511 – 1.0356599674090168

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0054575241831976	0.981300896638036 – 1.025779538880808

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9973555103077842	0.9729474862969293 – 0.9986107093239619

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9895963379109447	0.9548841059602649 – 1.0287800687285222

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0028067202414839	0.9733149389042176 – 1.0364563187268607

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0011050291676304	0.9775446246082573 – 1.0589875111507583

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.1025043680838673	0.9660265878877399 – 1.1978680879413723

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0519618239660657	1.006554307116105 – 1.1666666666666667

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.995463411620374	0.9778598743749026 – 1.0276817508382503

Cholesky factorization

256 × 256 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.01549232141547	1.0091252328428177 – 1.031882568924252

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

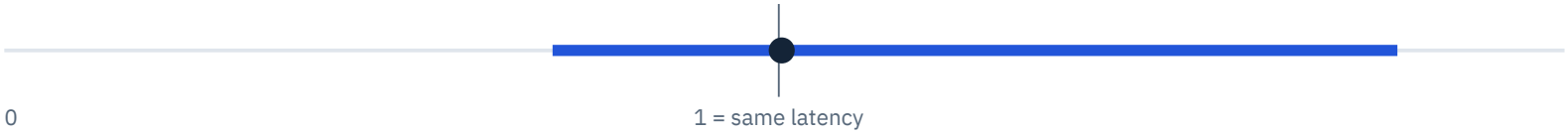
Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.00042872342826	0.9864813132222578 – 1.0167709594881083

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.003797530380243	0.7079180180333158 – 1.7986926866237212

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

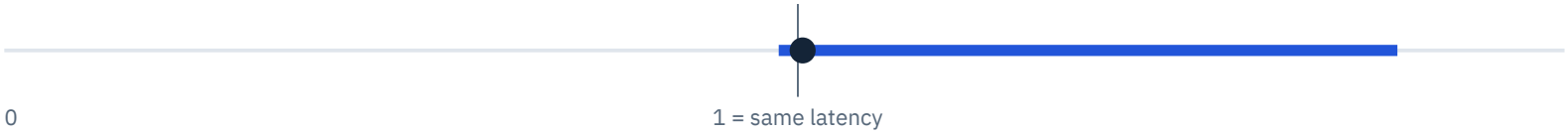
Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.965063468796831	0.9576436699021517 – 1.0025281357038003

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0060232840382972	0.9759132167367722 – 1.7554905490549055